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Integrated analysis of single-cell RNA-seq and bulk RNA-seq reveal macrophage subpopulation characteristics and the role of STAT1 in rheumatoid arthritis

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Abstract

Background

Rheumatoid arthritis (RA) represents a persistent systemic autoimmune disorder characterized by chronic inflammation and

significant disability. Macrophage heterogeneity plays a crucial role in the progression of RA, but the molecular characteristics and regulatory mechanisms of its subsets have not yet been fully elucidated.

Methods

In this study, we integrated single-cell RNA sequencing (scRNA-seq) data and bulk RNA sequencing (bulk RNA-seq) data, and employed multiple strategies for analysis and identification of the heterogeneity of macrophages in RA. We identified STAT1 as a key gene through a combined approach utilizing LASSO regression and random forest models. The differential expression of STAT1 was subsequently validated in an adjuvant-induced arthritis (AIA) rat model. Furthermore, functional experiments were performed to investigate the association of STAT1 with autophagy and ferroptosis pathways.

Results

Based on scRNA-seq data, a total of 26,923 cells were subjected to multi-dimensional analysis. The synovial tissues of RA mice exhibited a markedly elevated percentage of macrophages expressing Stat1. Enrichment analysis indicated that Stat1⁺ macrophages were concentrated in inflammatory pathways. Moreover, bulk RNA-seq and animal models further confirmed the upregulated expression of STAT1 in RA. Functional experiments demonstrated that STAT1 activation upregulated synovial LC3 and ACSL4, while downregulating p62 and GPX4. Treatment with fludarabine (Flu) reversed these changes, suggesting that STAT1 may contribute to RA pathogenesis by modulating autophagy and ferroptosis pathways.

Conclusions

This study elucidated the heterogeneity and functional characteristics of macrophage subsets in RA synovial tissue through scRNA-seq and bulk RNA sequencing, and suggested that STAT1 and its downstream signaling pathways may serve as potential therapeutic targets for RA.

Supplementary Information

The online version contains supplementary material available at [10.1186/s12967-025-07015-y](https://doi.org/10.1186/s12967-025-07015-y).

Keywords: Rheumatoid arthritis, Single-cell RNA sequencing, Bulk RNA sequencing, Macrophages, STAT1

Introduction

Rheumatoid arthritis (RA) is a chronic systemic autoimmune disease characterized by synovitis and inflammatory cell infiltration, often accompanied by progressive joint damage and extra articular symptoms [1, 2]. Research shows that RA is one of the diseases with the highest disability rates worldwide, affecting 0.24%–0.30% of the global population [3]. In addition, RA is also associated with early development of cardiovascular and interstitial lung diseases, as well as malignant tumors, which increase the risk of onset and premature death [4, 5]. The treatment of RA is challenging, usually requiring lifelong treatment and high treatment costs, which impose a heavy economic burden on patients and families. Therefore, revealing the pathogenesis and therapeutic targets of RA is central to its early prevention and treatment, and also represents the current research focus.

The pathogenesis of RA is complex, with genetic, environmental, and immune factors all involved in its occurrence and development. Among them, immune factors play a crucial role

throughout the entire process, especially in the early stages [6, 7].

The synovium of RA patients is infiltrated by various immune cells, including monocytes, T cells, macrophages, and dendritic cells (DCs), creating a complex inflammatory environment that drives joint inflammation [8]. Macrophages, the most representative immune cells in RA synovium, are the primary source of chemokines, cytokines, and degradation enzymes that drive joint inflammation and bone destruction. They also synergize with other immune cells to exacerbate arthritis [9]. For instance, macrophages drive the differentiation of CD4 + T cells into pro-inflammatory Th17 cells via cytokines such as IL-6 and IL-1 β , and in turn, Th17 and Th1 cells promote macrophage polarization toward the pro-inflammatory M1 phenotype through the secretion of IFN- γ , RANKL, and IL-22 [10]. Moreover, macrophages secrete soluble factors that transform fibroblasts into an aggressive phenotype, while fibroblast-derived products, along with chronic inflammatory signals, drive macrophage polarization toward M1, M2, or a hybrid pro-fibrotic phenotype [11].

M1 macrophages secrete proinflammatory mediators and express inducible nitric oxide synthase (iNOS), which promotes osteoclast-mediated bone resorption and joint destruction [12]. In contrast, M2 macrophages secrete anti-inflammatory factors such as TGF- β and IL-4/10, which inhibit inflammation [13]. The expansion of M1 macrophages in the synovium serves as an early biomarker for RA and correlates strongly with disease progression [14].

Regulating macrophage phenotypes has been shown to alleviate RA symptoms [15]. Thus, characterizing macrophage heterogeneity and identifying specific marker genes in synovial tissue are crucial for early diagnosis and targeted therapy in RA.

In recent decades, high-throughput transcriptome sequencing, particularly bulk RNA sequencing (bulk RNA-seq), has become a major tool in transcriptomics. While bulk RNA-seq provides tissue-wide transcriptional profiles, it only offers average expression levels and cannot reveal differences between individual cells [16].

In contrast, single-cell RNA sequencing (scRNA-seq) can detect subtle transcriptomic changes in individual cells, aiding in the identification of cellular heterogeneity, cell sources, functions, and new therapeutic targets [17, 18]. In this study, we integrated scRNA-seq and bulk RNA-seq to elucidate the distribution and molecular characteristics of RA-associated immune cell populations. We validated the expression of marker genes in synovial tissues and characterized intercellular interactions among distinct cell subpopulations, identifying potential therapeutic targets and disease-associated genes. Using an adjuvant-induced arthritis (AIA) rat model, we further validated the regulatory roles of these key genes in downstream signaling pathways.

Methods

Data collection

In this study, publicly available datasets related to RA and normal control tissues were retrieved from the Gene Expression Omnibus (GEO) database. The scRNA-seq data included two datasets: [GSE221704](#) and [GSE230145](#). Additionally, five bulk RNA sequencing datasets were collected: [GSE12021](#), [GSE55235](#), [GSE55457](#), [GSE77298](#), and [GSE89408](#). After filtering, a total of 213 RA samples and 63 healthy control (HC) samples were included for subsequent analysis. To ensure consistency, only genes commonly detected across all five bulk datasets were retained, resulting in a total of 4,232 gene symbols. All datasets were analyzed using R (Version 4.3.2), and the results were visualized with the ggplot2 R package (Version 3.3.5), unless otherwise specified.

Integration of scRNA-seq datasets

The Seurat (V5.0.1), standard workflow for scRNA-seq analysis was utilized to process and integrate the datasets [19]. Seurat

objects were created for each dataset, and quality control (QC) measures were implemented to exclude low-quality cells and doublets. Specifically, cells with fewer than 250 detected genes, mitochondrial gene content exceeding 10%, or sequencing depth below 500 were filtered out. In addition, DoubletFinder was applied to identify and remove doublets [20]. After quality control, a total of 26,923 cells from the respective datasets were retained for downstream analysis.

To correct for batch effects, variable genes were subjected to dimensionality reduction via principal component analysis (PCA), followed by batch effect correction and dataset integration using the Harmony algorithm [21]. Specifically, RunHarmony was applied with parameters: `group.by.vars = "orig.ident"`, `reduction.use = "pca"`, `theta = 2`, `lambda = 1`, `sigma = 0.1`, `nclust = 1`, and `early_stop = TRUE`, while other parameters were kept as defaults. The number of significant principal components was determined using an elbow plot based on the standard deviation of principal components, and the first 20 components showing clear inflection points were selected for subsequent analyses. Dimensionality reduction was performed using Seurat's RunUMAP and RunTSNE functions with parameters `reduction = "harmony"` and `dims = 1:20`; other parameters remained at their default values. A nearest-neighbor graph was constructed to identify clusters. Clustering resolution was optimized with the assistance of a clustering tree, and the final resolution was set to 0.1, enabling effective identification of distinct cell subpopulations.

Cell clusters were annotated based on the expression of canonical marker genes associated with specific cell types. Gene expression across clusters was visualized using dot plots, allowing for the identification of distinct cell populations and their corresponding molecular signatures.

Myeloid cell subtype analysis

Following the initial clustering, myeloid cells were extracted and subjected to re-clustering with dimensionality reduction techniques. To annotate the subtypes of myeloid cells, specific marker genes were used. The identified subtypes and their representative markers included Stat1⁺ macrophages (Stat1, Tgfb3), Fn1⁺ macrophages (Fn1, S100a4, Vcan), Il7r⁺ macrophages (Il7r, Ccr9, Flt3), Cd14⁺ monocytes (Cd14, S100a6), and C1q⁺ macrophages (C1qa, C1qc, C1qb).

Differentially expressed genes (DEGs) for myeloid cell subtypes were identified using the FindAllMarkers function in the Seurat package. The results were visualized with the scRNAtoolVis package to highlight subtype-specific expression patterns. Furthermore, DEGs between Stat1⁺ macrophages in the RA and control groups were identified using the FindMarkers function in Seurat, with thresholds set to $|\text{avg_log2FC}| > 1$ and adjusted p -value < 0.05 .

The identified DEGs were analyzed for functional enrichment to elucidate their biological significance. The gene names of the DEGs were mapped to gene IDs using the R package org.Mm.eg.db. Functional enrichment analyses included Gene Ontology (GO) analysis to identify enriched biological processes, molecular functions, and cellular components, as well as Kyoto Encyclopedia of Genes and Genomes (KEGG) analysis to detect significantly enriched signaling pathways. These analyses were conducted using the R package clusterProfiler [22], with a significance threshold of p -value < 0.05 and q -value < 0.05 . The findings were visualized using the R package ggplot2.

Pseudotime analysis of myeloid cells

To investigate the dynamic changes in myeloid cell states,

pseudotime analysis was performed using the Monocle3 package [23]. The normalized gene expression matrix and clustering information obtained from Seurat were converted into a Monocle3 object, and dimensionality reduction was conducted using the UMAP method to construct the trajectory. Cd14⁺ monocytes were selected as the starting point of differentiation, and cells were ordered along the pseudotime trajectory based on their transcriptional states. Key branch points were identified to reveal major transitions in cell differentiation or activation states. For the analysis of DEGs along the pseudotime trajectory, the graph_test function in Monocle3 was applied with thresholds of q_value < 0.01 and Morans_I > 0.25. The identified DEGs were clustered into four modules, and their expression patterns were visualized using heatmaps generated with the pheatmap package. Each module of DEGs was subjected to GO enrichment analysis using the clusterProfiler package. Significantly enriched pathways (p-value < 0.05) were identified and presented, providing insights into the functional roles of the gene modules during the dynamic transition of myeloid cells.

Cell-cell communication analysis

To compare cell-cell communication between the RA and control groups, intercellular signaling interactions were analyzed using the CellChat pipeline (version 1.6.1). A separate CellChat object was created for each group from the respective Seurat objects [24]. The CellChatDB database of mice was used as the reference. Communication probabilities were calculated using the truncated mean method. Afterward, cell-cell communication networks for each group were inferred and aggregated using default parameters. To assess differences in cell-cell communication between the RA and control groups, the total number of inferred interactions and the interaction strength were calculated and compared. Differential signaling pathways between the two groups were analyzed, and the relative contribution of specific signaling pathways (e.g., ITGAL-ITGB2) was evaluated. In

addition, targeted analyses were conducted to examine the interactions of Stat1⁺ macrophages with fibroblast subpopulations and T/NK cells, aiming to further elucidate their communication roles within the RA synovium.

Human bulk RNA sequencing analysis

We first performed preprocessing on the bulk RNA-seq datasets. Using the appropriate platform annotation files, we converted the probe expression matrix into a gene expression matrix. For genes associated with multiple probes, the average expression value across all probes was calculated as the gene's expression level. Subsequently, we merged the gene expression matrices to form a comprehensive gene expression matrix. To eliminate heterogeneity introduced by different experimental batches and platforms, batch effect correction was performed using the ComBat function in the sva package [25], and quantile normalization across datasets was applied using the normalizeBetweenArrays function in the limma package with default parameters to address cross-platform differences. After normalization, we obtained a merged, standardized gene expression matrix.

To identify DEGs between groups, differential expression analysis was conducted using the limma package [26], with filtering criteria set as adjusted P value (adj.PVal) < 0.05 and |log fold change (FC)| > 1. After obtaining DEGs, an intersection analysis was performed between the upregulated genes and the previously identified upregulated genes specific to Stat1⁺ macrophages (average log2FC > 1); the intersecting genes were used as the basis for subsequent analyses.

For these intersecting genes, we first conducted feature selection using LASSO regression implemented in the glmnet package [27]. Specifically, a LASSO model was fitted using glmnet where the

predictor variable consisted of expression levels of 4,232 genes, and the response variable was a binary disease status (healthy control, HC = 0; rheumatoid arthritis, RA = 1). The LASSO regularization parameter (λ) was optimized via 10-fold cross-validation, evaluating a geometric sequence of 100 λ values generated by glmnet's default algorithm. The optimal λ was selected based on minimizing the binomial deviance criterion. Through L1 regularization, gene features associated with RA were identified. Subsequently, gene importance was further assessed using the Boruta and randomForest packages [28]. Boruta was applied to refine the LASSO-selected variables by evaluating their importance scores, with only those confirmed as important retained for subsequent model construction. In the random forest analysis, confidence intervals for Mean Decrease Accuracy (MDA) were calculated as $MDA \pm 1.96 \times sd_MDA$, and permutation tests were performed to obtain *p*-values for selecting statistically significant features. Furthermore, we calculated the correlations between these selected genes and scores of multiple biological pathways, including ferroptosis, autophagy, and inflammation pathways, to explore their potential involvement in key pathogenic processes.

Animals and experimental design

All animal procedures were performed in accordance with the ethical guidelines approved by the Laboratory Animal Ethics Committee of Guangxi University of Chinese Medicine (No. 20240603-168). Male Sprague-Dawley (SD) rats, aged 6–8 weeks and weighing 200–250 g, were provided by Hunan SJA Laboratory Animal Co., Ltd. (Changsha, China). The rats were housed under specific pathogen-free (SPF) conditions with free access to food and water, at a controlled temperature (23 ± 2) °C and relative humidity (50 ± 5) %. Rats were randomly divided into three groups: normal control group (control), AIA model group (model), and AIA model treated with fludarabine group (Flu), with eight rats in each group.

In this study, we employed the AIA model, which is one of the commonly used animal models in RA research [29]. Except for the control group, arthritis was induced in all other groups as previously described [30]. On day 0 (designated as the day of AIA induction), rats received a single injection of 0.1 mL complete Freund's adjuvant (CFA) (10 mg/mL, Sigma-Aldrich, St. Louis, MO, USA) into the plantar region of the left hind paw. The success of the model induction can be determined by observing the redness and swelling of the joints in the limbs of rats, as well as the pathological changes, such as inflammatory cell infiltration, shown by HE staining [31].

Rats in the Flu group received daily intraperitoneal injections of fludarabine (20 mg/kg; Selleck, USA) at the same time each day from day 15 to day 20 [32], while control and model groups were administered equal volumes of saline. On day 21, all rats were euthanized and ankle joints along with synovial tissues were collected for subsequent analysis.

Arthritic assessment

Arthritis severity was evaluated on days 0, 7, 14, and 21 using the arthritis index (AI), following established methodology [33]. Concurrently, the hind paw volumes of the rats were measured with a plethysmometer (PV-200; TME Technology, Chengdu, China) at the same timepoints. Treatment efficacy was determined by comparing group means via double-blinded independent evaluation.

Histopathological assessment

The ankle joint tissues of rats were fixed in 4% paraformaldehyde for 48 h and then decalcified in 10% EDTA solution for 35 days. Subsequently, the tissues were sequentially dehydrated, clarified, and embedded in paraffin. The tissues were sectioned into 5 µm-

thick slices using a microtome. After dewaxing and rehydration, the sections were stained with hematoxylin-eosin (HE) solution, followed by sealing with neutral gum. The results were observed, photographed, and recorded under a microscope.

Rat joints were histologically scored (0–4) for inflammation severity: 0, no pathological change; 1, minimal periarticular inflammatory cell infiltrate; 2, mild inflammatory cell infiltrate; 3, moderate inflammatory cell infiltrate; 4, pronounced inflammatory cell accumulation [34].

Immunohistochemistry

Tissue sections were deparaffinized, rehydrated, and subjected to antigen retrieval for 10 min, followed by three PBS washes. Sections were blocked at room temperature for 10 min, washed three times with PBS, and then blocked with 5% BSA for 15 min to quench endogenous peroxidase activity. Sections were incubated overnight at 4 °C with primary antibodies against STAT1, LC3 and GPX4, then washed three times with PBS, incubated with secondary antibodies at room temperature for 1 h, and washed again three times with PBS. Sections were stained with DAB, counterstained with hematoxylin, dehydrated, and mounted. Images were captured under a microscope, and quantitative analysis was performed using Image-Pro Plus 6.0 (Media Cybernetics, USA).

Quantitative real-time PCR analysis

Total RNA was extracted from ankle joint synovial tissues using the TRIzol reagent (Invitrogen, Carlsbad, CA, USA) according to the manufacturer's instructions. The purity and concentration of the RNA were measured using a nucleic acid analyzer, and then the RNA was reverse-transcribed into cDNA. Quantitative real-time PCR was performed on the Light Cycler 96 system. The

reaction conditions were as follows: initial denaturation at 95 °C for 30s, followed by 40 cycles of denaturation at 95 °C for 5s, annealing at 60 °C for 30s, and extension at 72 °C for 20s. The relative expression levels of the STAT1 gene were analyzed using the $2^{-\Delta\Delta C_t}$ method with GAPDH as the internal reference gene. The primer sequences are shown in Table 1.

Table 1.

Primer sequences for quantitative real-time polymerase chain reaction

Gene	Primer sequence (5'-3')	Product/bp
STAT1	F: CTACTCCAGGCCAAAGGAAGC	133
	R: GAGGTTCTCAACAAGCCAGTC	
LC3	F: GCCTTCTCTAAAAGGGCTTCTACT	85
	R: TGGAGACCTCTGCTAGGCAA	
ACSL4	F: GAGGCAGCACCTTCGATCC	159
	R: AACGGAAGCAGCAGTAAGAGT	
GPX4	F: CCATTCCCGAGCCTTTCAAC	71
	R: GAACTCGTGGCTGTTGCTG	
P62	F: GAGTCGGGCATCGAGGTTG	99
	R: ATCTTCTGTGCCTGTGCTGG	
GAPDH	F: TCTCTGCTCCTCCCTGTTCT	95
	R: ATCCGTTACACCGACCTTC	

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Western blotting analysis

Frozen synovial tissues were powdered under liquid nitrogen and lysed in RIPA buffer on ice for 30 min, followed by centrifugation at 12,000 $\times$ g for 30 min at 4 °C. Protein concentration was assessed using a BCA assay (Solarbio, Beijing, China). Proteins were separated by 10% SDS-PAGE and transferred to PVDF membranes (Millipore Sigma, USA). Membranes were blocked with 5% skim milk for 1 h at room temperature and incubated overnight at 4 °C with primary antibodies against STAT1, LC3A/B, ACSL4/FACL4, GPX4, SQSTM1/p62 (1:1000; Affinity, Nanjing, China), and GAPDH (1:5000; Biosharp, Hefei, China). After TBST washing, membranes were incubated with HRP-conjugated goat anti-mouse IgG (1:5000; Servicebio, Wuhan, China) for 1 h. ECL substrate (Biosharp, Hefei, China) was applied for signal detection, followed by imaging. Band intensities were quantified using ImageJ software, with target protein expression normalized to GAPDH.

Statistical analysis

The data were analyzed using IBM SPSS Statistics 26.0, and the figures were created using GraphPad Prism 8.0. Numerical data are expressed as the mean $\pm$ standard deviation (mean $\pm$ SD). Comparisons between two groups were performed using the t-test, with $P < 0.05$ considered statistically significant.

Results

Single-cell transcriptional profiling of control and RA mice

In this study, we analyzed eight tissue samples from the [GSE221704](#) and [GSE230145](#) datasets, including four from control

mice and four from RA mice. After quality control, normalization, data integration, and principal component analysis, a total of 26,923 cells were retained for further analysis. Single-cell clusters were annotated using canonical marker genes and visualized through UMAP dimensionality reduction, ultimately resolving distinct cell populations: fibroblasts (*Col3a1*, *Col1a1*, *Igfbp6*), granulocytes (*Clec4d*, *Il1b*, *Acod1*), neutrophils (*Ltf*, *Ifitm6*, *Ngp*), stem cells (*Cenpm*, *Hist1h1a*), Pro-B cells (*Rag1*, *Atp1b1*, *Arl5c*), B cells (*Ms4a1*, *Igkc2*, *Fcmmr*), endothelial cells (*Cdh5*, *Gpihbp1*, *Cldn5*), osteoblasts (*Alpl*, *Bglap*, *Sp7*), monocytes (*Mycl*, *S1pr5*, *Ms4a4a*), naïve B cells (*Cd79b*, *Cd74*), macrophages (*C1qa*, *Mrc1*, *Cd163*), NK cells (*Txk*, *Klrk1*, *Fasl*), erythrocytes (*Car2*, *Car1*, *Klf1*), pericytes (*Notch3*, *Tagln*, *Acta2*), mesenchymal stem cells (MSCs) (*Myod1*, *Pax7*, *Myf5*), and chondrocytes (*Chad*, *Fxyd2*, *Sox9*) (Fig. 1A, E). UMAP visualization revealed that the spatial distribution of certain cell clusters, such as fibroblasts and macrophages, in the RA group was significantly different from that in the control group (Fig. 1B). Analysis of cell proportions showed a significant increase in the proportions of fibroblasts and macrophages in the RA group (Fig. 1C). Additionally, the DEGs between RA and control groups in macrophages totaled 206, second only to chondrocytes with 224, highlighting their significant role in RA (Fig. 1D).

Fig. 1.



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Overview of single-cell transcriptional profiling in control and RA mice. **A** UMAP plots of cell clustering and annotation results for single-cell datasets. **B** Comparison of UMAP clustering plots between control and RA mice. **C** Comparison of cell type proportions between control and RA mice. Statistical significance is indicated as follows: *** $p < 0.001$. **D** DEGs between RA and control groups across cell

types. E Bubble plots illustrated the expression of marker genes for 16 cell types

Single-cell transcriptomic atlas of monocyte-macrophages via scRNA-seq

In this study, we identified 1,432 monocyte-macrophages from single-cell transcriptomic data for clustering analysis and annotation of cell subpopulations. Using unsupervised dimensionality reduction and clustering, we categorized five distinct subclusters of monocyte-macrophages within a two-dimensional t-SNE plane and employed the FindAllMarkers function in the Seurat package to identify marker genes for each subcluster. Based on the marker genes, we further annotated the monocyte-macrophage populations as follows: Stat1⁺ macrophages (Stat1, Tgfbr3), C1q⁺ macrophages (C1qa, C1qc, C1qb), Fn1⁺ macrophages (Fn1, S100a4, Vcan), Il7r⁺ macrophages (Il7r, Ccr9, Flt3), and Cd14⁺ monocytes (Cd14) (Fig. 2A, C, and D).

Fig. 2.



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Single-cell transcriptome atlas of monocyte-macrophages in scRNA-seq. **A** UMAP plot of monocyte-macrophages in scRNA-seq, clustered into five distinct clusters. **B** Comparison of UMAP clustering plots between control and RA mice. **C** Comparison of the relative expression of marker genes in different cell types. **D** Bubble plot depicting differential expression of Stat1 in five types of macrophages. **E** Comparison of the proportions of macrophage subsets between control and RA mice. Statistical significance is indicated as follows: $**p < 0.01$. **F** The UMAP plot highlighted the expression of marker genes for five

macrophage subpopulations

The t-SNE visualization results indicate that macrophages in the control mice exhibited distinct subpopulation distribution characteristics, while the distribution pattern of macrophage subpopulations was significantly altered in RA mice (Fig. 2B).

The comparison of cell proportions further revealed the differences in monocytes and macrophages between the two groups. The results showed that the proportions of Stat1⁺ macrophages and Cd14⁺ monocytes were significantly increased in the RA mice compared with the control mice, with the most significant increase observed in Stat1⁺ macrophages. In contrast, the proportions of C1q⁺ macrophages, Fn1⁺ macrophages, and Il7r⁺ macrophages were decreased (Fig. 2E, Table S1).

FeaturePlot further demonstrated the expression of marker genes in each cell cluster (Fig. 2F).

Differential enrichment analysis of macrophage-associated pathways

Volcano plot analysis revealed DEGs in cell clusters, including Stat1⁺ macrophages, Fn1⁺ macrophages, Il7r⁺ macrophages, Cd14⁺ monocyte, and C1q⁺ macrophages (Fig. 3A). GO and KEGG enrichment analyses of Stat1⁺ macrophages marker genes identified significant associations with ferroptosis, autophagy (animal), and cell cycle pathways (Fig. 3B-D). These pathways are known to involve STAT1-mediated mechanisms in macrophages, including lipid peroxidation regulation [35], autophagosome formation [36], and cell cycle suppression [37].

Fig. 3.



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Differential enrichment analysis of macrophages. **A** Volcano plot of intercellular differences in the overall cell subpopulations. Numbers above each cluster indicate total DEGs. **B** KEGG enrichment analysis of DEGs in macrophage cells. **C** GO functional enrichment map of DEGs among macrophage cells. **D** GO enrichment analysis of biological processes, cellular components, and molecular functions. **E** Scatter plot of differential analysis among macrophage groups. **F** Bar plot of DEGs enriched in macrophage groups

GO biological process analysis further demonstrated that marker genes for Stat1⁺ macrophages were significantly enriched in processes such as macroautophagy, cellular component disassembly, and lipid catabolic process, which further supports the role of these cells in autophagy and lipid metabolism. GO analysis of cellular components and molecular functions revealed that marker genes for Stat1⁺ macrophages are significantly enriched in functions such as microtubule, autophagosome, and GTP binding, suggesting their potential involvement in cytoskeleton dynamics and signal transduction processes.

Additionally, there were significant differences in gene expression between the control group and the RA mice group in Stat1⁺ macrophages (Fig. 3E). GO pathway enrichment analysis (Fig. 3F) revealed that pathways such as antigen processing and presentation and RNA splicing were significantly activated in Stat1⁺ macrophages, which are closely related to immune regulation and modification of gene expression. For instance, the phosphorylation of STAT1 can initiate the expression of interferon regulatory factor (IRF)-1, which in turn promotes the expression of MHC class I molecules, thereby enhancing antigen presentation on the cell surface. This process plays a critical role in immune responses and disease progression [38].

Pseudotime analysis of macrophages

Visualization of single-cell transcriptomic data by UMAP dimensionality reduction revealed distinct clustering patterns among different macrophage subpopulations (Fig. 4A). Pseudotime analysis revealed the differentiation trajectories of macrophage subpopulations (Fig. 4B). Of note, the cell populations of the healthy and RA mice exhibited significant distribution differences in the UMAP space (Fig. 4C), indicating that disease status may influence the differentiation characteristics

of macrophages. The pseudotime gene heatmap revealed sequential activation of different pathways during myeloid cell differentiation. Notably, the Stat1⁺ macrophage cluster exhibited significantly increased activity in iron-related processes, including cellular response to iron ions, intracellular iron ion homeostasis, and iron ion transmembrane transport. GO enrichment analysis further confirmed the important roles of these cell clusters in key pathways such as the humoral immune response, in addition to the aforementioned iron-related processes (Fig. 4D).

Fig. 4.



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Pseudotime analysis of specific subpopulations. **A** Distribution of cell clusters in the UMAP space. **B** Distribution of cell pseudotime in the UMAP space. **C** Distribution of cells from control and RA mice in the UMAP space. **D** Pseudotime differential gene distribution heatmap and enrichment profile

Analysis of macrophage-associated intercellular interactions

To investigate the communication characteristics between macrophages and other cell types, we employed the CellChat software to analyze the ligand-receptor pairs and molecular interactions among different cells. The results demonstrated that ligand-receptor-mediated cell interactions were prominently involved in multiple signaling pathways and exhibited marked differences in RA mice (Fig. 5A). The cell-cell interaction network revealed the connections in terms of communication quantity and strength between different cell types in control and RA mice. In RA mice, interactions between specific cell types were significantly enhanced, such as increased interactions between Stat1⁺ macrophages and other cell types including C1q⁺ macrophages, Cd14⁺ monocyte, and Fn1⁺ macrophages (Fig. 5B). Differential analysis identified ligand-receptor pairs that were specifically upregulated in RA mice, among which the ITGAL-ITGB2 interaction axis exhibited significant activation (Fig. 5C). We further focused on the differences in interaction strength of ligand-receptor pairs between Stat1⁺ macrophages and other cells. Notably, fibroblasts are one of the key cells that influence RA, and their crosstalk with other cells may be an important factor in the progression of RA. Interestingly, the interaction between Stat1⁺ macrophages and fibroblasts showed significant differences only in the Itgb2-Icam1 ligand-receptor pairs ($P < 0.01$), suggesting that the differences in these ligand-receptor pairs may be one of the important factors contributing to the pathological changes in RA (Fig. 5D). Further contribution analysis revealed that the Itgb2-Icam1 ligand-receptor pair had the highest contribution in the ITGAL-ITGB2 interaction axis, indicating that this ligand-receptor pair may play a crucial role in the pathogenesis of RA (Fig. 5E). Gene expression analysis further confirmed that the expression of integrin family genes (Itga1, Itgb2) and immune regulatory genes (Cd226) was significantly upregulated in the RA group (Fig. 5F). These genes may

participate in the pathological process of RA by regulating inflammatory responses and infiltration of immune cells.

Fig. 5.



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Analysis of cell communication. **A** Comparison of the number and strength of cell communication between control and RA mice. **B** Circle plot of intercellular interaction

numbers between control and RA mice. **C** Differences in relative information flow of signaling pathways between control and RA mice. **D** Comparison of changes in signaling pathways involved in different intercellular interactions between control and RA mice. **E** The relative contributions of different molecular interaction pairs in the ITGAL-ITGB2 signaling pathway. **F** Expression distribution of marker genes in different cell types between control and RA mice

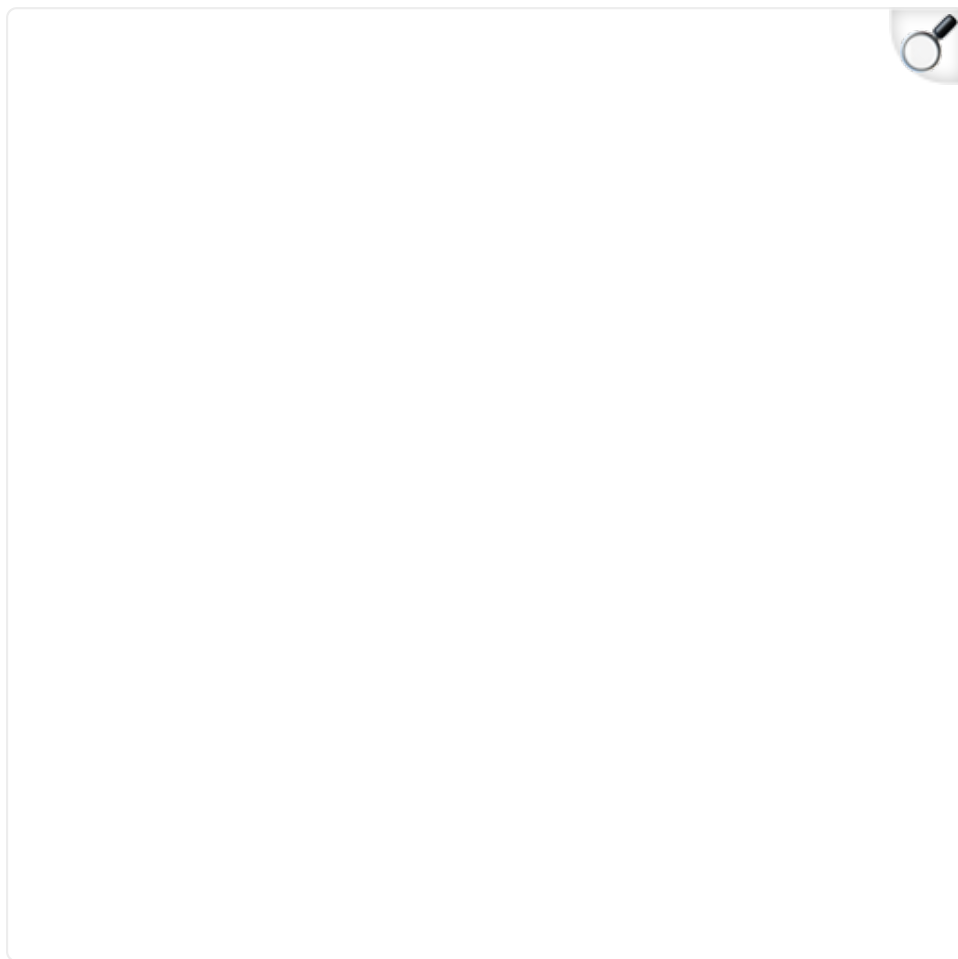
Furthermore, we characterized the intercellular crosstalk between Stat1⁺ macrophages and other cellular subpopulations, with a particular focus on fibroblasts and T cell subsets. Our results showed that Stat1⁺ macrophages exerted significant effects on fibroblast subpopulations through multiple signaling pathways, including ITGAL–ITGB2, SEMA4, TGFβ, and VCAM (Figure [S1A](#)). Detailed analysis of cellular roles within the ITGAL–ITGB2 pathway revealed that Stat1⁺ macrophages primarily functioned as key Senders and Influencers, whereas Cd34⁺ fibroblasts and Cilp⁺ fibroblasts within the fibroblast compartment were identified as the main Receivers of this signal (Figure [S1B](#)). In T cell subsets, we similarly identified several critical intercellular interaction pathways, including CCL, COMPLEMENT, ITGAL–ITGB2, and CD48 (Figure [S1C](#)). Among these, CCL represents a pivotal pathway mediating immune cell chemotaxis. Role analysis within the CCL pathway indicated that Stat1⁺ macrophages again served predominantly as Senders and Influencers, with the primary chemotactic targets being Cd8⁺ T cells and NK cells. Notably, Stat1⁺ macrophages exhibited minimal interaction with Cd4⁺ T cells via the CCL pathway (Figure [S1D](#)).

Validation of the role of STAT1 in RA through bulk RNA-seq analysis

We performed differential gene expression analysis between

control and RA patients using the limma package and identified 1,434 significantly upregulated genes in RA patients (Fig. 6A). These DEGs were then intersected with marker genes of Stat1⁺ macrophages, revealing 65 overlapping genes (Fig. 6B). Subsequent feature selection using LASSO regression with L1 regularization further refined these candidates, yielding 30 RA-associated key genes (Fig. 6C).

Fig. 6.



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Bulk RNA-seq analysis of control and RA patients. **A** Volcano plot depicting differential gene expression between the control and RA patients. **B** Venn diagram showing the

overlap between marker genes of Stat1⁺ macrophages and differentially upregulated genes between groups. **C** Lasso regression was employed to screen the intersecting genes. **D** Box plot of the importance scores for different genes in the Boruta model. **E** Bar plot of the importance scores of different genes in the random forest model. **F** Correlation between STAT1 expression and gene set scores (ferroptosis, autophagy and inflammation pathways). Spearman's correlation coefficient (ρ) and corresponding p -values are shown

To evaluate gene importance, these 30 candidate genes were assessed using the Boruta model. We identified nine key genes—GPR65, STAT1, PTTG1, C1QA, SHTN1, FCGR3B, TMEM176B, CD74, and SLC7A7—that were consistently prioritized (Fig. 6D). Notably, STAT1 demonstrated high importance scores in both models: (1) in Boruta, a meanImp value of 8.25 was observed (Table S2), $P < 0.05$; (2) in random forest, MeanDecreaseGini = 1.67 and MeanDecreaseAccuracy = 0.0661 (95% CI: 0.0503–0.0818, $P < 0.05$) (Table S3) (Fig. 6E).

Furthermore, correlation analyses revealed that STAT1 expression was significantly associated with ferroptosis (Spearman's $\rho = 0.402$, $p = 3.79 \times 10^{-12}$), autophagy (Spearman's $\rho = 0.246$, $p = 3.47 \times 10^{-5}$), and inflammation (Spearman's $\rho = 0.435$, $p = 3.55 \times 10^{-14}$), suggesting a potential involvement of STAT1 in these biological processes in RA (Fig. 6F).

Validation of the impact of STAT1 on autophagy and ferroptosis pathways in the AIA rat model

To elucidate the expression characteristics of STAT1 in RA pathogenesis and its association with autophagy and ferroptosis pathways, we employed the AIA rat model for functional

validation. The results demonstrated that, compared with the control group, CFA-injected rats exhibited marked paw swelling, with significantly increased paw volumes and arthritis index scores (both $P < 0.01$). Flu treatment significantly ameliorated these symptoms, showing markedly reduced paw volumes and arthritis scores in the Flu group compared to the model group ($P < 0.01$) (Fig. [7A-C](#)).

Fig. 7.



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Validation of the impact of STAT1 on autophagy and ferroptosis pathways in the AIA rat model. **A** The appearance of rat paws in each group. **B** Paw volume of rats in each group. **C** Arthritis scores of the rats in each group. **D** Histopathological analysis of the joint tissue of rats. **E** Inflammatory scores of joint tissues. **F–G** Representative images and quantitative analysis of GPX4, LC3, and STAT1

immunohistochemical staining in the synovium (magnification, x400). **H** RT-qPCR analysis of p62, LC3, GPX4, ACSL4 and STAT1 in synovial tissue. **I–J** Western blot examination revealed the expression of the proteins STAT1, ACSL4, p62, GPX4, and LC3 in synovial tissue. Data are shown as means $\pm$ SD. *. $P < 0.05$ vs. control group; **. $P < 0.01$ vs. control group; #. $P < 0.05$ vs. model group; ##. $P < 0.01$ vs. model group

Histopathological assessment of ankle joint sections from the rats in each group revealed no significant hyperplasia, necrosis, or inflammatory cell infiltration in the synovial cells (blue arrows) or sub-synovial connective tissue in the control group. In contrast, the model group exhibited mild synovial cell hyperplasia (yellow arrows), synovial thickening, substantial connective tissue hyperplasia (green arrows), numerous neovascular formations (red arrows), and significant inflammatory cell infiltration (purple arrows). The Flu group showed a marked reduction in these pathological changes compared to the model group (Fig. [2D](#)). Pathological scoring indicated that the synovial inflammation score was significantly higher in the model group than in the control group ($P < 0.01$), and was significantly lower in the Flu group compared to the model group ($P < 0.01$) (Fig. [2E](#)).

Immunohistochemical analysis revealed that compared with the control group, the expression levels of STAT1 and LC3 proteins in the synovial tissue of the model group rats were significantly increased (both $P < 0.01$), while the expression of GPX4 protein was markedly decreased ($P < 0.01$). In comparison with the model group, the expression of STAT1 and LC3 proteins in the synovial tissue of the Flu group rats was notably reduced ($P < 0.05$ and $P < 0.01$, respectively), and the expression of GPX4 protein was significantly elevated ($P < 0.05$) (Fig. [2F-G](#)).

Quantitative real-time PCR analysis revealed that, compared with the control group, the mRNA expression levels of STAT1, LC3, and ACSL4 were significantly elevated (all $P < 0.01$) in the synovial tissue of rats in the model group, whereas the mRNA expression levels of p62 and GPX4 were significantly decreased ($P < 0.01$ and $P < 0.05$, respectively). In comparison with the model group, the mRNA expression levels of STAT1, LC3 (both $P < 0.01$), and ACSL4 ($P < 0.05$) were significantly reduced in the Flu group, while the mRNA expression levels of p62 and GPX4 were significantly increased (both $P < 0.05$) (Fig. 7H). Western blot results were concordant with the qRT-PCR findings (Fig. 7I-J). Collectively, these data suggest that STAT1 may contribute to RA pathogenesis by regulating autophagy and ferroptosis-related pathways.

Discussion

RA is a complex and heterogeneous disease, with significant challenges in diagnosis and treatment, and substantial individual differences in treatment resistance and disease progression [39, 40]. Over the past decade, biological heterogeneity has been recognized as a major driver of variable clinical outcomes and treatment responses in RA [41, 42]. Macrophages, one of the most heterogeneous and functionally diverse cell populations in RA pathology, can switch between the pro-inflammatory “M1-like” phenotype and the immunomodulatory “M2-like” phenotype [43]. Given the close link between macrophage phenotype switching and inflammatory progression in RA, a thorough analysis of their heterogeneity and causal relationship with RA pathogenesis can provide a vital theoretical basis for developing new therapeutic approaches.

In this study, we analyzed 26,923 cells from synovial tissues of 4 healthy and 4 RA mice, identifying 1,432 macrophages, including 506 Stat1 + cells, 397 Fn1 + cells, 273 Il7r + cells, 68 C1q + cells,

and 188 Cd14⁺ monocytes. RA mice exhibited significantly altered macrophage compositions and functional states compared to healthy controls, with increased proportions of Stat1⁺ macrophages and Cd14⁺ monocytes, and decreased proportions of Fn1⁺, Il7r⁺, and C1q⁺ macrophages. STAT1, a key transcription factor for IFN- γ signaling, is associated with M1-polarized pro-inflammatory macrophages [44] and is overexpressed in synovial tissues of RA animal models [45], consistent with our findings. In human RA, scRNA-seq identified a STAT1-high Mj2 subpopulation [46], and SPP1⁺ STAT1⁺ macrophages correlated with JAK inhibitor response [47]. Cd14⁺ monocytes in RA mice display hyperinflammatory and hypermetabolic characteristics [48], resembling the M1 phenotype. The C1q⁺ subset exhibits both M1 and M2 characteristics, suggesting a dual role in immune regulation [49]. These findings indicate that the imbalance of macrophage subsets in RA synovium is associated with alterations in both pro-inflammatory and anti-inflammatory phenotypes.

KEGG and GO enrichment analyses showed that Stat1⁺ macrophages are highly enriched in pathways related to ferroptosis, autophagy, and lipid metabolism. STAT1 is a significant factor in autoimmune diseases like RA [50, 51]. Autophagy, an intracellular clearance mechanism that degrades damaged organelles and protein aggregates via lysosomes, maintains cellular homeostasis. Its abnormal levels are closely associated with RA progression and pathology [52]. The activation of STAT1 can induce the accumulation of LC3-II and p62 via an IRF1-dependent pathway, thereby triggering aberrant autophagy [36], and can also drive the release of TNF- α and IL-1 β from macrophages through the STAT1-IRF1-TLR3 signaling pathway [53]. Additionally, autophagy directly contributes to bone destruction by regulating osteoclast differentiation [54].

Ferroptosis is an iron-dependent form of regulated cell death characterized by iron deposition, lipid peroxidation, glutathione

(GSH) depletion, and glutathione peroxidase 4 (GPX4) inactivation, all of which are closely related to the pathological progression of RA [55]. Activation of STAT1 can transcriptionally repress the expression of the cystine-glutamate antiporter (System Xc⁻), thereby reducing intracellular GSH levels and promoting ferroptosis [35]. Lipid metabolic dysregulation has been reported in approximately 60% of RA patients and correlates with disease activity [56]. This metabolic disturbance exacerbates inflammation by enhancing the pro-inflammatory functions of macrophages [57]. Collectively, these findings suggest that autophagy, ferroptosis, and lipid metabolism pathways may collectively mediate the pro-inflammatory effects of Stat1⁺ macrophages.

Enrichment analysis of DEGs further indicated that pathways such as antigen processing and presentation and RNA splicing were significantly enriched in Stat1⁺ macrophages of RA mice. As crucial antigen-presenting cells (APCs), macrophages are capable of taking up, processing antigens, and presenting them to T cells, thereby initiating immune responses [58]. In RA, aberrant antigen presentation by Stat1⁺ macrophages may activate self-reactive T cells, leading to chronic inflammation and joint damage [59]. Dysregulated RNA splicing can upregulate pro-inflammatory cytokines or immune-related proteins, thereby exacerbating inflammation [60]. Additionally, RNA splicing alterations may affect the expression of cell-surface molecules, such as MHC and co-stimulatory molecules, influencing antigen presentation and immune cell interactions [61]. These findings suggest that targeting Stat1⁺ macrophages or their downstream pathways may be a potential therapeutic strategy.

Pseudotemporal trajectory analysis delineated the differentiation pathway of myeloid cells in RA, revealing a stepwise progression from monocytes to multiple macrophage subsets. This likely reflects the dynamic evolution and functional differentiation of myeloid cells in the RA inflammatory microenvironment. During

disease progression, Stat1⁺ macrophages were positioned at the terminus of this trajectory, indicating terminal differentiation into pro-inflammatory cells. Their sustained activation may contribute to tissue damage and disease chronicity [62]. GO enrichment analysis revealed pathways activated at distinct cell-fate stages, such as “regulation of lymphocyte proliferation,” “defense response to bacterium,” and “cellular response to iron ion,” which were significantly enriched in Stat1⁺ macrophages. Notably, the enrichment of the “cellular response to iron ion” pathway aligns with findings implicating ferroptosis in RA pathogenesis [63]. In RA, Stat1⁺ macrophages promote ferroptosis by secreting chemokines and cytokines to recruit and activate other immune cells, and may also undergo ferroptosis via lipid peroxidation and ROS generation [63]. Thus, Stat1⁺ macrophages have a dual role in RA as both inflammatory drivers and regulators of ferroptosis.

The complex pathogenesis of RA involves alterations in individual cell types and remodeling of intercellular interaction networks [64]. Our analysis revealed enhanced interactions between Stat1⁺ macrophages and other cell types, particularly Cd14⁺ monocytes and C1q⁺ macrophages, in RA mice. These interactions likely reflect activation of pro-inflammatory cytokine networks and immune cell coordination during RA inflammatory responses [65, 66]. Specific cell populations associated with RA, such as THY1⁺ fibroblasts, exacerbate inflammatory arthritis through interactions with macrophages and endothelial cells [65]. We characterized the crosstalk between Stat1⁺ macrophages and fibroblasts/T cell subsets [46, 47]. Stat1⁺ macrophages significantly influenced fibroblasts via pathways like ITGAL–ITGB2 and SEMA4, with Cd34⁺ and Cilp⁺ fibroblasts as main receivers. In T cells, Stat1⁺ macrophages primarily interacted with Cd8⁺ T cells and NK cells via the CCL pathway, with minimal interaction with Cd4⁺ T cells. These interactions may drive distinct roles of T cell subsets in RA.

In this study, we observed significant upregulation of ITGAL and ITGB2 expression in RA mice, indicating increased levels of the

LFA-1 (lymphocyte function-associated antigen 1) complex, which may activate the LFA-1 signaling pathway. Ligand-receptor interaction analysis identified ITGAL-ITGB2 as a key mediator of macrophage–T cell/fibroblast crosstalk, promoting inflammatory cell synergy and consistent with its role in immune adhesion and activation [66]. Previous studies show elevated LFA-1 expression in high-affinity states on synovial fluid T cells from RA patients, suggesting synovial microenvironment-driven T cell activation [67]. Moreover, LFA-1 activates downstream Src kinase via ICAM-1 binding, enhancing IFN- γ -induced STAT1 phosphorylation [67]. Thus, aberrant LFA-1 and STAT1 activation in RA may represent potential therapeutic targets.

To investigate the role of STAT1 in RA, we performed bulk RNA-seq analysis, identifying 1,434 upregulated genes in RA patients. LASSO regression further refined these to 30 key genes, with STAT1 showing high importance in both Boruta and random forest models. Correlation analyses showed significant associations between STAT1 expression and ferroptosis, autophagy, and inflammation. These findings prompted animal experiments, which confirmed elevated STAT1 expression in AIA rat synovial tissue compared with controls, consistent with previous reports [45]. To elucidate the role of STAT1 in RA pathogenesis via autophagy and ferroptosis, we treated AIA rats with Flu and assessed changes in autophagy markers (LC3, p62) and ferroptosis markers (GPX4, ACSL4).

During autophagy, cytosolic LC3-I conjugates with phosphatidylethanolamine to form membrane-bound LC3-II, thereby promoting autophagosome formation. The LC3-II/LC3-I ratio is a well-established autophagy marker [68]. Conversely, p62 accumulation inversely correlates with autophagic flux due to its selective degradation [69]. In AIA rats, we observed joint swelling, synovial inflammation, elevated LC3-II/LC3-I ratios, and reduced p62 levels, consistent with autophagy activation [70]. Flu treatment attenuated these effects. In ferroptosis regulation, GPX4

is a key inhibitor; its impairment depletes glutathione (GSH), leading to lipid peroxide (LPO) and reactive oxygen species (ROS) accumulation, which disrupts redox homeostasis and induces ferroptosis [71]. Conversely, ACSL4 promotes ferroptosis by catalyzing the synthesis of polyunsaturated fatty acid-containing lipids, thereby increasing lipid peroxidation [72]. In our AIA rat model, ACSL4 was upregulated and GPX4 downregulated, indicating ferroptosis, consistent with prior reports [73]. Flu intervention reversed this trend. In summary, these results suggest that STAT1 may contribute to RA pathogenesis by modulating autophagy and ferroptosis pathways.

This study unveiled the molecular characteristics of STAT1⁺ macrophages at the single-cell level and explored the potential mechanisms by which STAT1 regulates autophagy and ferroptosis in the pathogenesis of RA through transcriptomic analysis and animal experiments. However, there are some limitations to this study. Firstly, the prognostic genes identified through analysis of public databases are based on relatively small sample sizes, and future validation in larger cohorts is necessary. Secondly, the mechanisms of action of STAT1 have only been verified in the AIA model, and further confirmation using clinical samples and methods such as immunohistochemistry is required. Additionally, the specific targets of Flu and its interaction mechanisms with STAT1 still require further investigation.

Conclusion

Our multi-omics analysis reveals macrophage heterogeneity and pro-inflammatory mechanisms in RA, highlighting STAT1-driven intercellular networks as key pathogenic contributors. Functional studies establish STAT1 activation as a therapeutic target by modulating autophagy and ferroptosis. These findings provide mechanistic insights and a rationale for targeting STAT1 signaling in RA treatment.

Supplementary Information

Below is the link to the electronic supplementary material.

[Supplementary Material 1](#) (13.6KB, xlsx)

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Author contributions

LLY and XLH conceived and designed the study, XLH, QYT and GF analyzed data, LLY and YNB wrote the manuscript, YZP and AH revised the manuscript and supervised the work. All authors have read and approved the final manuscript.

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Data availability

The datasets analyzed in this study were obtained from the Gene Expression Omnibus (GEO) database. The raw data of this article

will be available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

All the authors declare that they have no competing interests.

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Associated Data

This section collects any data citations, data availability statements, or supplementary materials included in this article.

Supplementary Materials

[Supplementary Material 1](#) (13.6KB, xlsx)

Data Availability Statement

The datasets analyzed in this study were obtained from the Gene Expression Omnibus (GEO) database. The raw data of this article will be available from the corresponding author upon reasonable request.

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